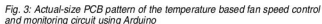


Fig. 2: Screenshot of the source code on Arduino IDE



Fan speed control technique. A low-frequency pulse-width modulation (PWM) signal, usually in the range of about 30Hz, whose duty cycle is varied to

adjust the fan's speed is used. An inexpensive, single, small pass transistor can be used here. It is efficient

One disadvantage of this approach, however, is that it can make the fan noisy because of the pulsed nature of the signal. The PWM waveform's sharp edges cause the fan's mechanical structure to move (like a badly-designed loudspeaker), which can easily be audible.

An actual-size, single-side PCB for the temperature based fan speed control and monitoring circuit is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on the PCB. CON2 and CON3 are used to connect Board1 (Arduino UNO board) through external connectors. A 12V battery is used to drive the 12V DC-operated fan.

Software for the automatic temperature controller and monitor circuit is written in Arduino programming language. Arduino Uno is programmed using Arduino IDE software. ATmega328P on Arduino Uno comes with a pre-programmed bootloader that allows users to upload a new code to it without using an external hardware programmer.

Connect Arduino board to the PC and select the correct COM port in Arduino IDE. Compile the program (sketch). Then select the correct board from Tools→Board menu in Arduino IDE and upload the sketch (abfc.ino) to Arduino through standard USB port. **EFY**

